

SIMPLE IMPACT STUDIES — SAMPLE DRAFT

Charleston County Sample



Traffic Impact Study

Prepared September 2, 2026 · 32.8560, -79.9750

Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Study evaluates the proposed Charleston County Sample within Charleston (SC) MSA, South Carolina (Charleston region (Charleston / Berkeley / Dorchester)), in accordance with SCDOT's Access and Roadside Management Standards (ARMS, 2008 edition, updated July 8, 2025) Chapter 6 and Traffic Engineering Guideline TG-21. In South Carolina the TIS is an element of the SCDOT encroachment-permit process: the study scope must be established with the District Traffic Engineer (DTE) before analysis begins, and the DTE makes the final mitigation determination.

12	680	0	2
INTERSECTIONS	PM PEAK TRIPS	AT LOS E/F	LOS DROPS

1.0 INTRODUCTION

Project name	Charleston County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	32.8560°, -79.9750°
Build-out year	2027
Controlling jurisdiction	Charleston region (Charleston / Berkeley / Dorchester)
MPO / study	CHATS (Charleston Area Transportation Study), staffed by BCDCOG
Programmed projects source	SCDOT FFY 2024–2033 STIP, as amended (e-STIP)
Submittal channel	SCDOT EPPS (Encroachment Permit Processing System) via the District Permit Engineer

1.1 TIS Applicability

City of Charleston TIS Preparation Guide (June 2021): a TIS is required when a development includes ANY of — a drive-through window; more than 6 fuel dispensers; more than 10,000 sf of non-residential building coverage; a site of 5 acres or more; a driveway-spacing variance request; a restaurant over 4,000 sf GFA; or single/two-family projects totaling 45 or more units. The statewide ARMS 100 peak-hour-trip trigger applies on SCDOT facilities.

The proposed development's estimated peak-hour generation of 680 trips meets or exceeds the ARMS 100 peak-hour-trip threshold; a TIS is required for the encroachment permit.

2.0 PROJECT TRAFFIC

ARMS Chapter 6 calls for trip generation per the latest edition of the ITE Trip Generation Manual, with internal-capture, pass-by, and transit reductions justified to the DTE. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal.

SCDOT District 6 caps for Charleston-area studies: internal capture limited to 20% of the lesser of entering/exiting trips; pass-by limited to 10% of adjacent street traffic. The credits applied in this screening analysis must be checked against these caps at submittal.

Daily rate applied	80.00 trips per ksf × 85 ksf
AM peak rate applied	3.20 trips per ksf × 85 ksf
PM peak rate applied	8.00 trips per ksf × 85 ksf

Rate source	SANDAG 2002 “(Not So) Brief Guide of Vehicular Traffic Generation Rates for the San Diego Region” — Community Shopping Center (daily 80/ksf; PM pass-by 30%)
Daily trips (gross)	6,800
AM peak hour	166 in / 106 out
PM peak hour	326 in / 354 out
Counts standard	AM and PM weekday peaks minimum; counts ≤ 12 months old, school in session, seasonally adjusted

2.1 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^{1.1} \cdot d_{site})$; mass basis: intersection through-volume (AADT $\times$ K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector		Share of project trips			
Northeast (NNE + ENE)		14%			
Southeast (ESE + SSE)		18%			
Southwest (SSW + WSW)		29%			
Northwest (WNW + NNW)		39%			

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Rivers Avenue & Reynolds Avenue	SSE	0.17	801	0.1	11.40%
McMillan Avenue & Spruill Avenue	NNW	0.32	801	0.1	10.76%
Rivers Avenue & McMillan Avenue	WNW	0.43	801	0.1	10.36%
Cosgrove Avenue & Azalea Drive	SSW	0.67	846	0.1	10.10%
Cosgrove Avenue & Rivers Avenue	WSW	0.11	657	0.1	9.59%
Cosgrove Avenue & Spruill Avenue	NNW	0.23	657	0.1	9.13%
Cosgrove Avenue & North Hobson Avenue	NNE	0.62	700	0.1	8.51%
Dorchester Road & Rivers Avenue	WNW	0.24	612	0.1	8.48%
Dorchester Road & Meeting Street Road	WSW	0.40	612	0.1	8.00%
Spruill Avenue & North Carolina Avenue	ESE	0.60	549	0.1	6.71%
Reynolds Avenue & Spruill Avenue	ENE	0.28	423	0.1	5.78%
Dorchester Road & South Allen Drive	WSW	0.71	100	0.0	1.18%

12 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

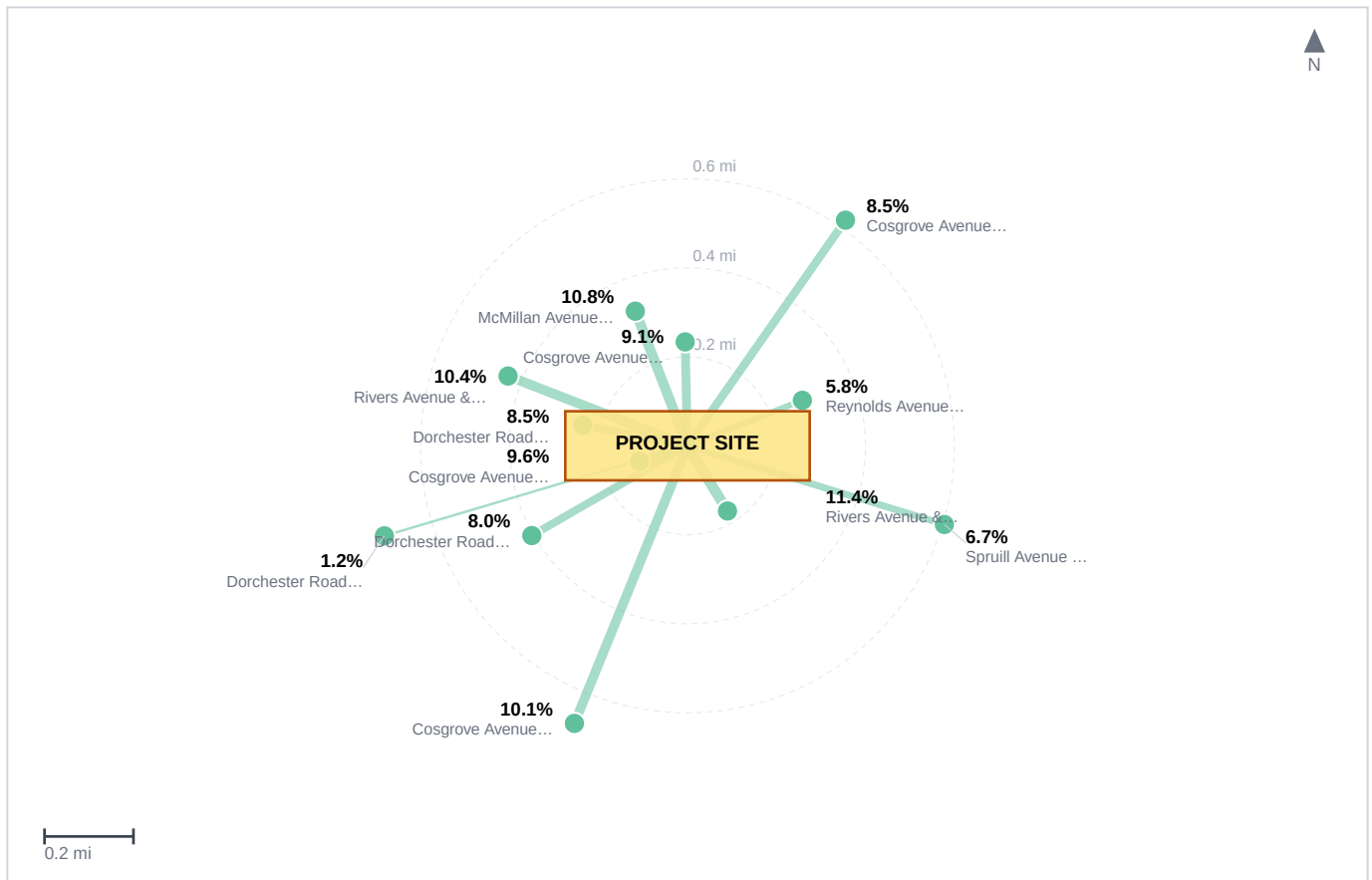
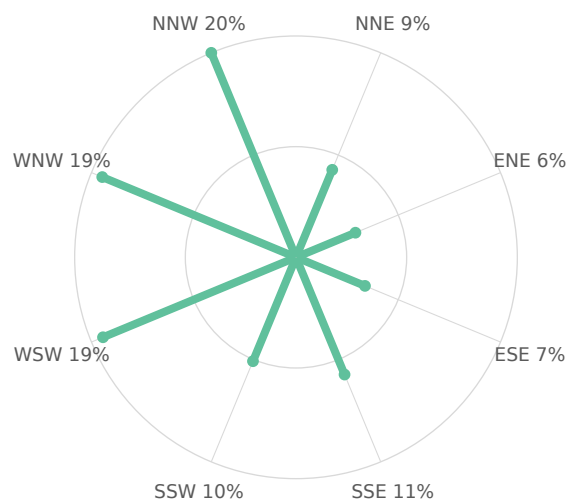


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

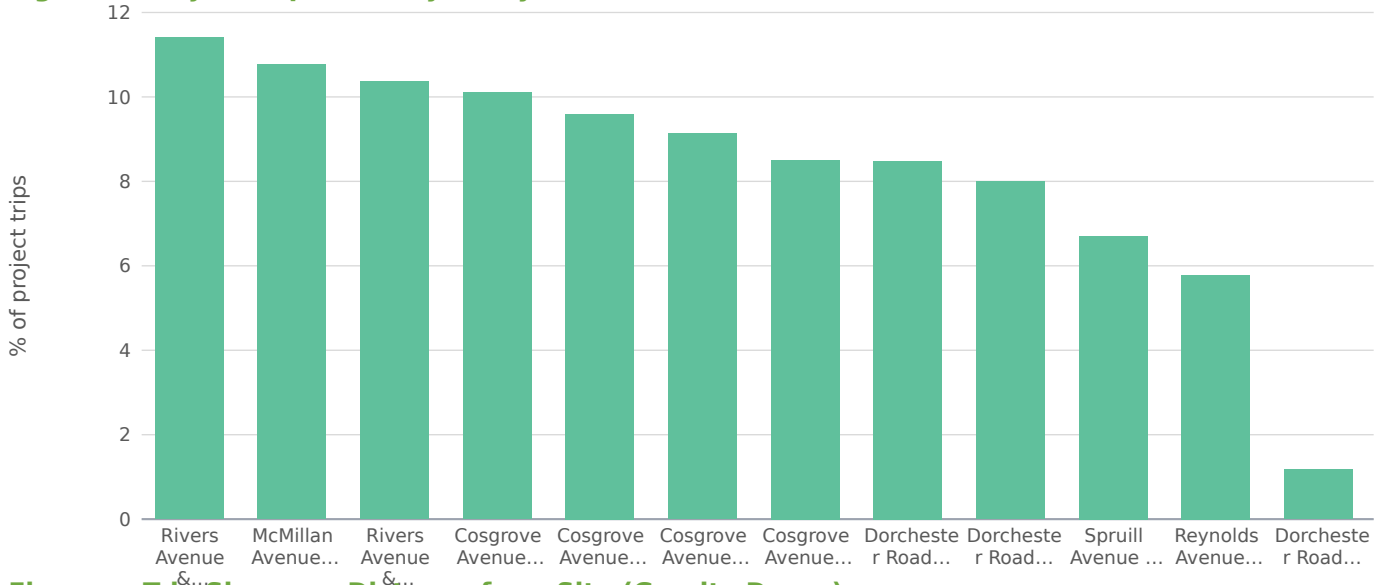
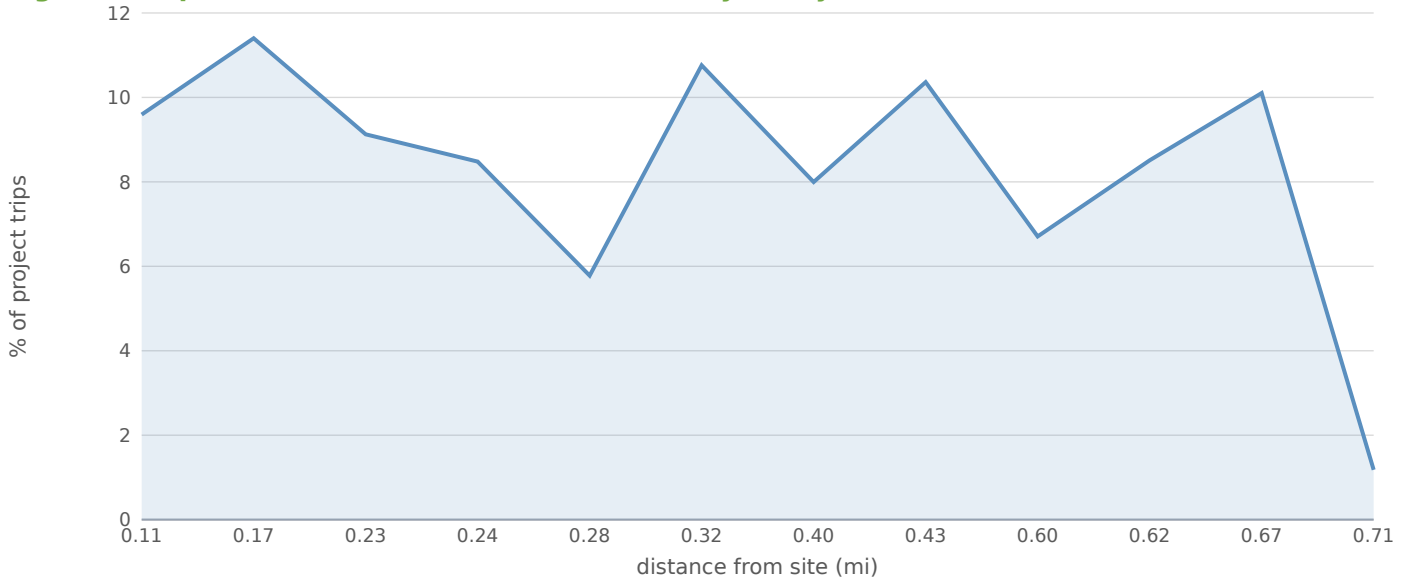


Figure — Trip Share vs. Distance from Site (Gravity Decay)



2.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Cosgrove Avenue & Rivers Avenue	WSW	0.11	166	312	22.5%
Rivers Avenue & Reynolds Avenue	SSE	0.17	55	105	7.6%
Cosgrove Avenue & Spruill Avenue	NNW	0.23	72	137	9.9%
Dorchester Road & Rivers Avenue	WNW	0.24	136	247	17.8%
Reynolds Avenue & Spruill Avenue	ENE	0.28	58	110	7.9%
McMillan Avenue & Spruill Avenue	NNW	0.32	46	87	6.3%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Dorchester Road & Meeting Street Road	WSW	0.40	44	82	5.9%
Rivers Avenue & McMillan Avenue	WNW	0.43	110	207	14.9%
Spruill Avenue & North Carolina Avenue	ESE	0.60	11	21	1.5%
Cosgrove Avenue & North Hobson Avenue	NNE	0.62	11	20	1.4%
Cosgrove Avenue & Azalea Drive	SSW	0.67	10	14	1.0%
Dorchester Road & South Allen Drive	WSW	0.71	17	43	3.1%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

3.0 TRAFFIC VOLUME DEVELOPMENT AND CAPACITY ANALYSIS

Scenarios per ARMS: existing year, build-out no-build, and build-out build. Capacity analysis computes signalized control delay by the openly-published Webster/Akcelik method with LOS reported for all approaches and movements; coordinated signal systems are analyzed as a system.

3.1 TG-21 Level of Service Standard — LOS C

Per SCDOT Traffic Engineering Guideline TG-21 (Mitigation of Traffic Impacts), the acceptable level of service for the design (peak) hour is LOS C or better for all roadway types statewide, applied in lieu of locally preferred thresholds. Where the baseline already operates at or below LOS C, the baseline LOS must be maintained or improved; where the baseline is LOS F in a congested urban area, mitigation is at the DTE's determination. Note: this is a stricter standard than the LOS D convention applied in several neighboring states.

No.	Intersection	Exist.	No-Build	Build LOS/Delay	TG-21 (LOS C) check
1	Cosgrove Avenue & Rivers Avenue	B	B	C / 20.6s	meets LOS C
2	Rivers Avenue & Reynolds Avenue	B	B	B / 20.0s	meets LOS C
3	Cosgrove Avenue & Spruill Avenue	B	B	B / 18.8s	meets LOS C
4	Dorchester Road & Rivers Avenue	B	B	B / 19.4s	meets LOS C
5	Reynolds Avenue & Spruill Avenue	B	B	B / 16.7s	meets LOS C
6	McMillan Avenue & Spruill Avenue	B	B	B / 19.8s	meets LOS C
7	Dorchester Road & Meeting Street Road	B	B	B / 17.9s	meets LOS C
8	Rivers Avenue & McMillan Avenue	B	B	C / 21.1s	meets LOS C
9	Spruill Avenue & North Carolina Avenue	B	B	B / 17.0s	meets LOS C
10	Cosgrove Avenue & North Hobson Avenue	B	B	B / 18.2s	meets LOS C
11	Cosgrove Avenue & Azalea Drive	B	B	B / 19.5s	meets LOS C
12	Dorchester Road & South Allen Drive	B	B	B / 14.3s	meets LOS C

4.0 FINDINGS, ACCESS MANAGEMENT, AND SIGNAL WARRANTS

2 intersections project a LOS drop and 0 operate at LOS E or F under Build conditions; mitigation alternatives (turn lanes with storage per ARMS Ch. 5 and the SCDOT Roadway Design Manual Ch. 9, signalization, or operational improvements) are developed with the DTE, who makes the final

determination.

Access management: ARMS requires demonstration that the fewest driveways necessary serve the site, with sight-distance verification at each access point. Signal proposals require a signal warrant analysis per the MUTCD. Both are site-plan-dependent and are prepared at submittal.

5.0 ARMS TECHNICAL COMPLETENESS CHECKLIST

SCDOT returns incomplete studies unreviewed. The ARMS completeness checklist is tracked below against this screening draft; unmet items are produced at formal submittal.

Checklist item	Status in this draft
SC PE stamp and signature	At submittal (screening draft is unsealed)
Introduction + executive summary	Included
Existing conditions (classification, peaks, LOS, volumes)	Included (screening level)
Trip generation / distribution / LOS analysis	Included (public-data rates, tagged)
Crash data / speeds / sight distance	At submittal (site-specific)
Mitigation w/ turn-lane storage lengths	Indicated per TG-21 table; design at submittal
Figures (vicinity, site plan, volumes, lane configs, LOS)	Volume/LOS figures included; site-plan figures at submittal
Tables (trip gen, LOS existing/background/build/mitigated)	Included (screening level)
Technical appendix (capacity reports, warrants)	Capacity worksheets appended; Synchro/warrants at submittal

This TIS is to be prepared under the direct charge of and sealed by a registered South Carolina Professional Engineer with expertise in traffic engineering (ARMS), with seal, signature across the seal face, and date per S.C. Code §40-22-270 and Regs. Ch. 49, and the firm's Certificate of Authorization. Electronic sealing is accepted in current practice.

SCREENING DISCLOSURE: this document is a screening-level draft prepared before DTE scoping. Scope, growth rates, background developments, and mitigation are established with the District Traffic Engineer; a submittal-grade analysis supersedes this draft.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Cosgrove Avenue & Rivers Avenue	0.11	0.3	312	22.5%
Dorchester Road & Rivers Avenue	0.24	0.6	247	17.8%
Rivers Avenue & McMillan Avenue	0.43	1.0	207	14.9%
Cosgrove Avenue & Spruill Avenue	0.23	0.6	137	9.9%
Reynolds Avenue & Spruill Avenue	0.28	0.7	110	7.9%
Rivers Avenue & Reynolds Avenue	0.17	0.4	105	7.6%
McMillan Avenue & Spruill Avenue	0.32	0.8	87	6.3%
Dorchester Road & Meeting Street Road	0.40	1.0	82	5.9%
Dorchester Road & South Allen Drive	0.71	1.7	43	3.1%
Spruill Avenue & North Carolina Avenue	0.60	1.4	21	1.5%
Cosgrove Avenue & North Hobson Avenue	0.62	1.5	20	1.4%
Cosgrove Avenue & Azalea Drive	0.67	1.6	14	1.0%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 92% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 8% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.29. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	2.23	260	0.29
Principal arterial	1.20	210	0.20
Minor arterial	0.51	40	0.17

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 12 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 205 network junctions, max imbalance 0.00 veh. 7 study intersection(s) carry path-derived movements; 5 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

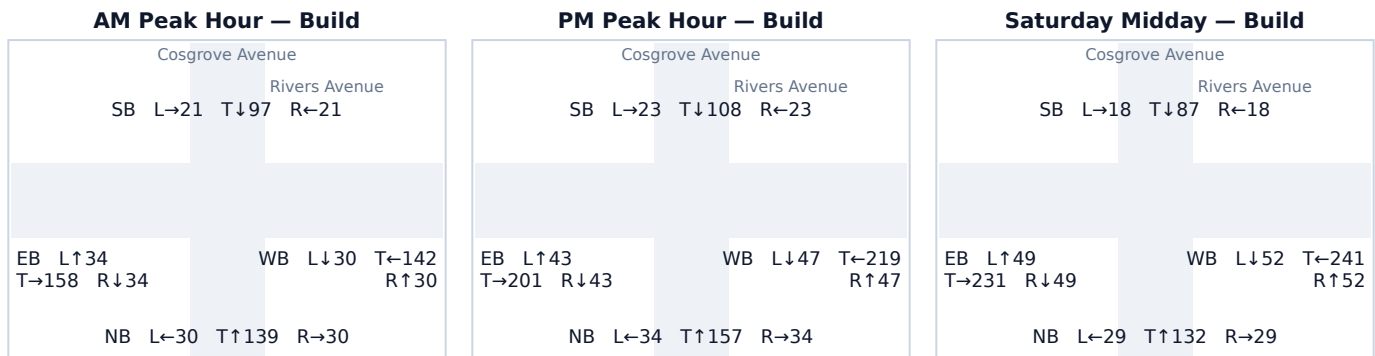
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Cosgrove Avenue & Rivers Avenue

Signal ID	charleston-sc-301
Location	NW Charleston (SC) · 0.11 mi from site
Added PM peak trips	312
Existing / No-Build (PM)	LOS B · 17.6 s/veh · v/c 0.37
Build (PM)	LOS C · 20.6 s/veh · v/c 0.54
Δ control delay	+3.0 s/veh (LOS change)
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	211	14	225	0.26 → 0.28	16.2 → 16.4	B → B	146
SB	154	0	154	0.19 → 0.19	15.4 → 15.4	B → B	96
EB	147	139	287	0.18 → 0.35	15.3 → 17.4	B → B	193
WB	155	159	313	0.19 → 0.39	15.4 → 17.9	B → B	215

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	159	51%
EB Through	139	45%
NB Right	14	4%

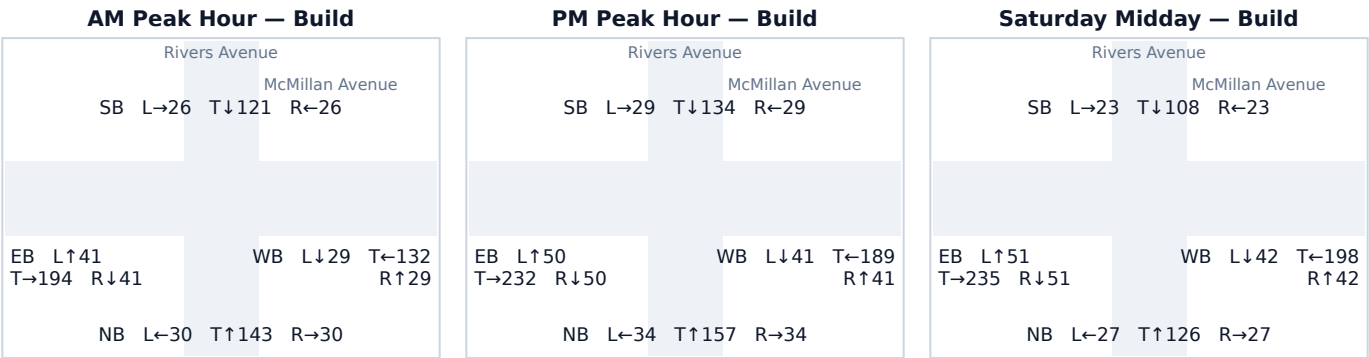
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Rivers Avenue & McMillan Avenue

Signal ID	charleston-sc-303
Location	NW Charleston (SC) · 0.43 mi from site
Added PM peak trips	207
Existing / No-Build (PM)	LOS B · 18.9 s/veh · v/c 0.45
Build (PM)	LOS C · 21.1 s/veh · v/c 0.57
Δ control delay	+2.2 s/veh (LOS change)
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	225	0	225	0.28 → 0.28	16.4 → 16.4	B → B	146
SB	192	0	192	0.24 → 0.24	15.9 → 15.9	B → B	122
EB	232	99	332	0.29 → 0.41	16.5 → 18.2	B → B	231
WB	164	108	271	0.20 → 0.34	15.5 → 17.1	B → B	181

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	108	52%
EB Through	99	48%

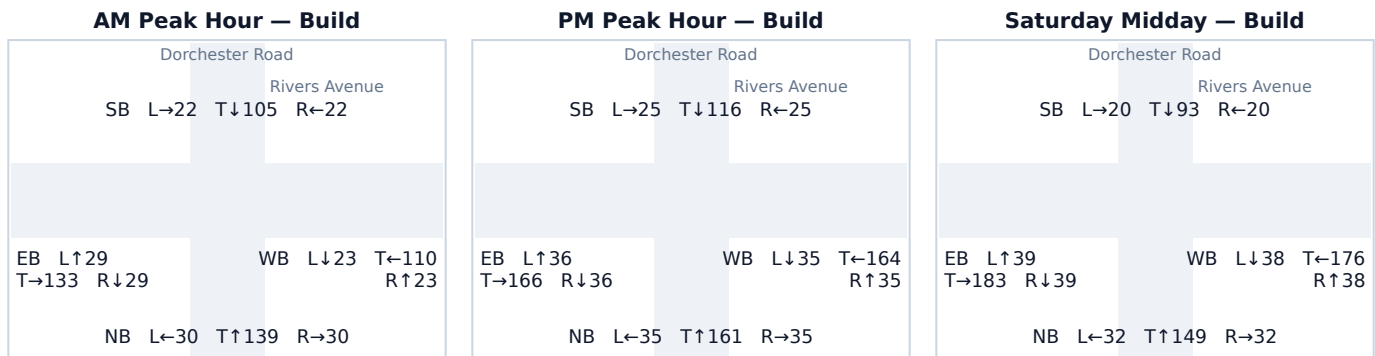
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Dorchester Road & Rivers Avenue

Signal ID	charleston-sc-304
Location	NW Charleston (SC) · 0.24 mi from site
Added PM peak trips	247
Existing / No-Build (PM)	LOS B · 17.3 s/veh · v/c 0.35
Build (PM)	LOS B · 19.4 s/veh · v/c 0.48
Δ control delay	+2.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	191	39	231	0.24 → 0.28	15.9 → 16.5	B → B	150
SB	166	0	166	0.20 → 0.20	15.6 → 15.6	B → B	104
EB	138	100	238	0.17 → 0.29	15.2 → 16.6	B → B	155
WB	126	108	234	0.16 → 0.29	15.0 → 16.5	B → B	153

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	108	44%
EB Through	100	40%
NB Right	39	16%

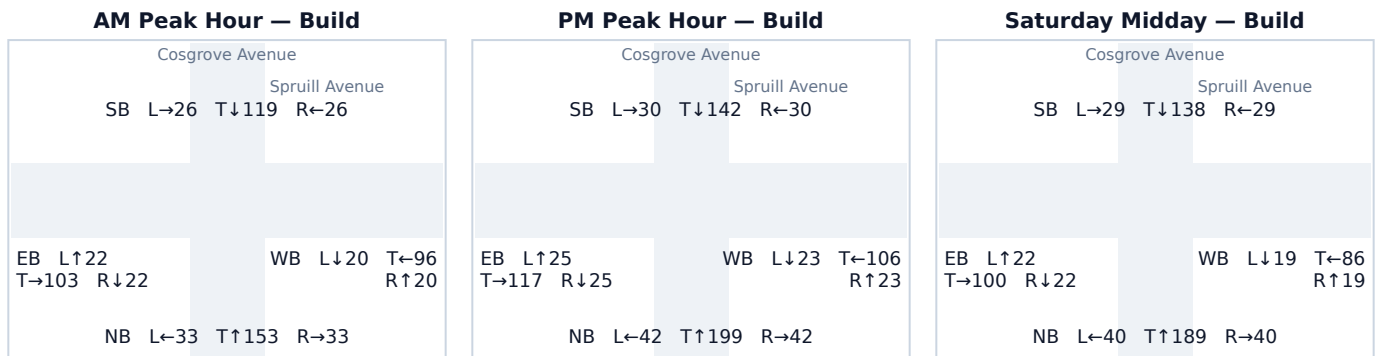
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Cosgrove Avenue & Spruill Avenue

Signal ID	charleston-sc-78
Location	NW Charleston (SC) · 0.23 mi from site
Added PM peak trips	137
Existing / No-Build (PM)	LOS B · 17.6 s/veh · v/c 0.37
Build (PM)	LOS B · 18.8 s/veh · v/c 0.45
Δ control delay	+1.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	212	71	283	0.26 → 0.35	16.2 → 17.3	B → B	190
SB	154	48	202	0.19 → 0.25	15.4 → 16.1	B → B	129
EB	153	14	167	0.19 → 0.21	15.4 → 15.6	B → B	105
WB	148	4	152	0.18 → 0.19	15.3 → 15.4	B → B	94

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	52	38%
SB Through	48	35%
NB Left	15	11%
EB Right	14	10%
NB Right	4	3%
WB Left	4	3%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

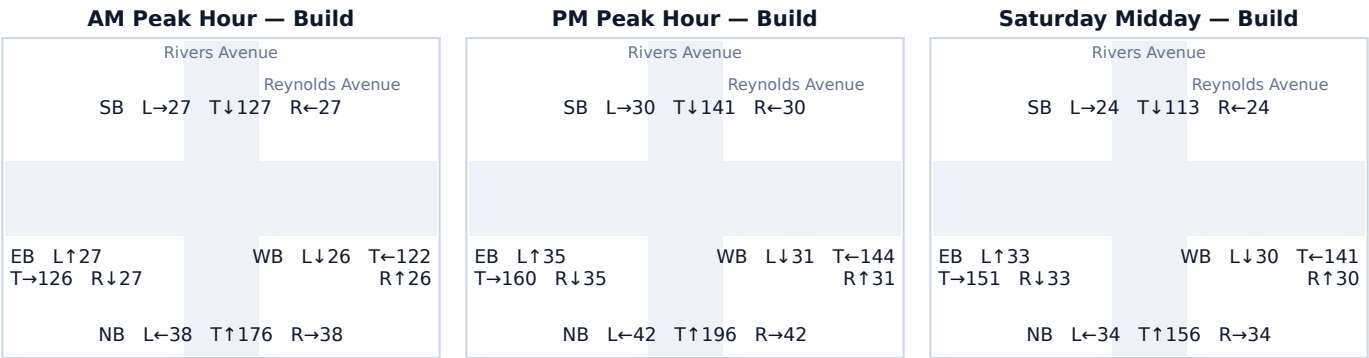
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into

the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Rivers Avenue & Reynolds Avenue

Signal ID	charleston-sc-147
Location	Central Charleston (SC) · 0.17 mi from site
Added PM peak trips	105
Existing / No-Build (PM)	LOS B · 18.9 s/veh · v/c 0.45
Build (PM)	LOS B · 20.0 s/veh · v/c 0.51
Δ control delay	+1.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	280	0	280	0.35 → 0.35	17.3 → 17.3	B → B	188
SB	201	0	201	0.25 → 0.25	16.1 → 16.1	B → B	129
EB	176	55	230	0.22 → 0.28	15.7 → 16.5	B → B	150
WB	156	50	206	0.19 → 0.25	15.4 → 16.1	B → B	132

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	55	52%
WB Through	50	48%

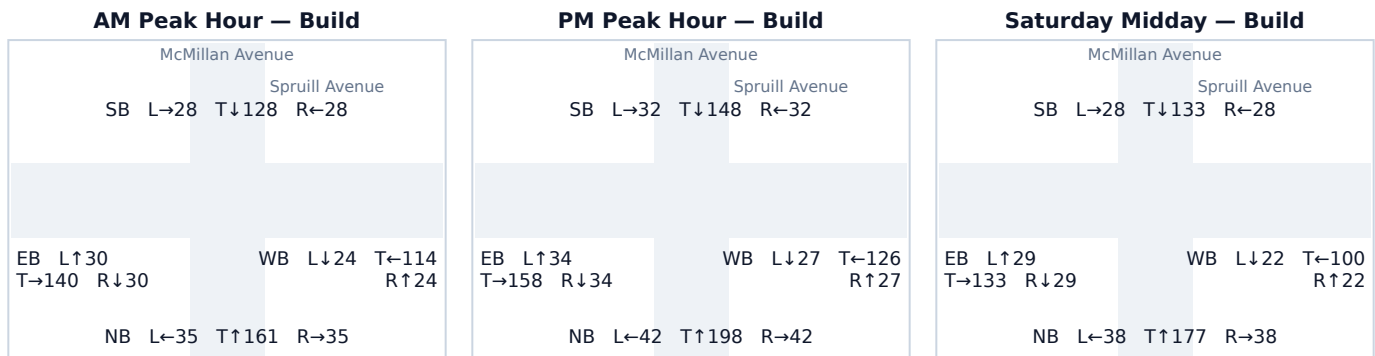
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 McMillan Avenue & Spruill Avenue

Signal ID	charleston-sc-77
Location	NW Charleston (SC) · 0.32 mi from site
Added PM peak trips	87
Existing / No-Build (PM)	LOS B · 18.9 s/veh · v/c 0.45
Build (PM)	LOS B · 19.8 s/veh · v/c 0.50
Δ control delay	+0.9 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	237	45	282	0.29 → 0.35	16.6 → 17.3	B → B	190
SB	184	28	212	0.23 → 0.26	15.8 → 16.2	B → B	136
EB	212	14	226	0.26 → 0.28	16.2 → 16.4	B → B	147
WB	179	0	180	0.22 → 0.22	15.8 → 15.8	B → B	113

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	30	34%
SB Through	28	32%
NB Left	15	17%
EB Right	14	16%

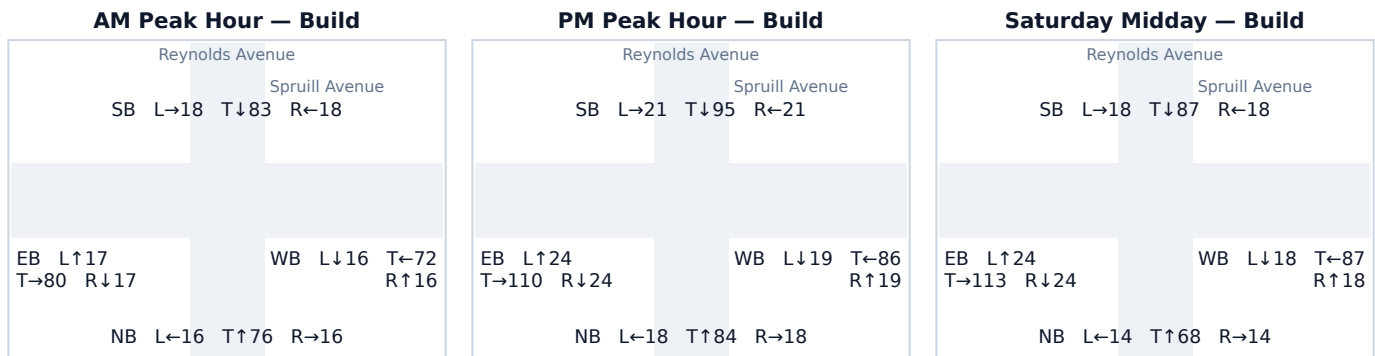
Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Reynolds Avenue & Spruill Avenue

Signal ID	charleston-sc-79
Location	NW Charleston (SC) · 0.28 mi from site
Added PM peak trips	110
Existing / No-Build (PM)	LOS B · 15.9 s/veh · v/c 0.24
Build (PM)	LOS B · 16.7 s/veh · v/c 0.30
Δ control delay	+0.8 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	119	1	120	0.15 → 0.15	15.0 → 15.0	B → B	73
SB	118	19	137	0.15 → 0.17	14.9 → 15.2	B → B	84
EB	101	57	158	0.12 → 0.20	14.7 → 15.5	B → B	98
WB	91	33	124	0.11 → 0.15	14.6 → 15.0	B → B	76

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	36	33%
WB Through	33	30%
EB Left	20	18%
SB Right	19	17%
EB Right	1	1%
NB Left	1	1%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

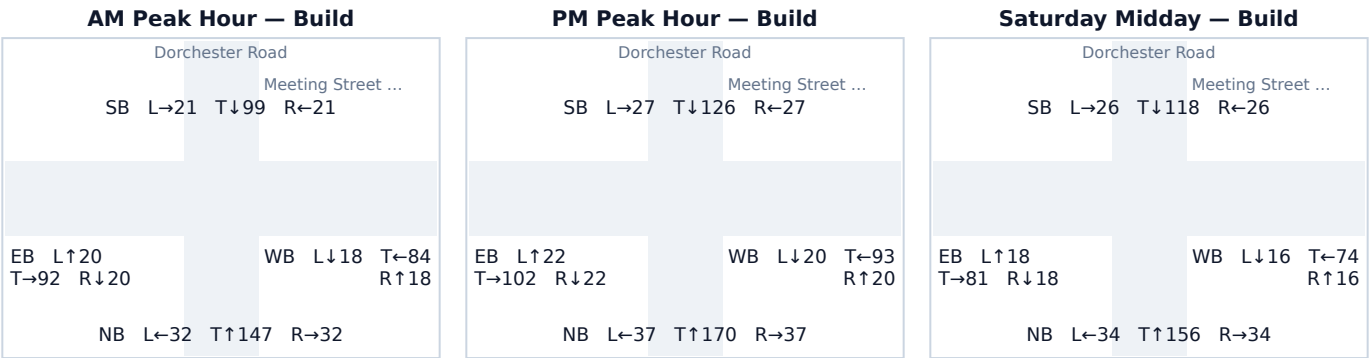
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into

the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Dorchester Road & Meeting Street Road

Signal ID	charleston-sc-362
Location	NW Charleston (SC) · 0.40 mi from site
Added PM peak trips	82
Existing / No-Build (PM)	LOS B · 17.3 s/veh · v/c 0.35
Build (PM)	LOS B · 17.9 s/veh · v/c 0.39
Δ control delay	+0.6 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	205	39	244	0.25 → 0.30	16.1 → 16.7	B → B	160
SB	137	43	180	0.17 → 0.22	15.2 → 15.8	B → B	114
EB	146	0	146	0.18 → 0.18	15.3 → 15.3	B → B	90
WB	133	0	133	0.16 → 0.16	15.1 → 15.1	B → B	81

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	43	52%
NB Through	39	48%

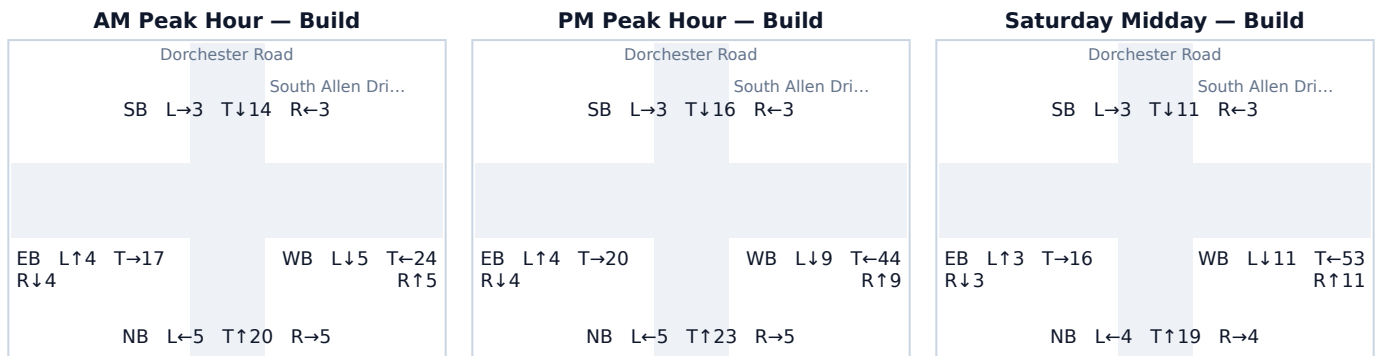
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Dorchester Road & South Allen Drive

Signal ID	charleston-sc-743
Location	NW Charleston (SC) · 0.71 mi from site
Added PM peak trips	43
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.3 s/veh · v/c 0.08
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	33	0	33	0.04 → 0.04	14.0 → 14.0	B → B	19
SB	22	0	22	0.03 → 0.03	13.8 → 13.8	B → B	12
EB	28	0	28	0.03 → 0.03	13.9 → 13.9	B → B	16
WB	19	43	62	0.02 → 0.08	13.8 → 14.3	B → B	36

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	43	100%

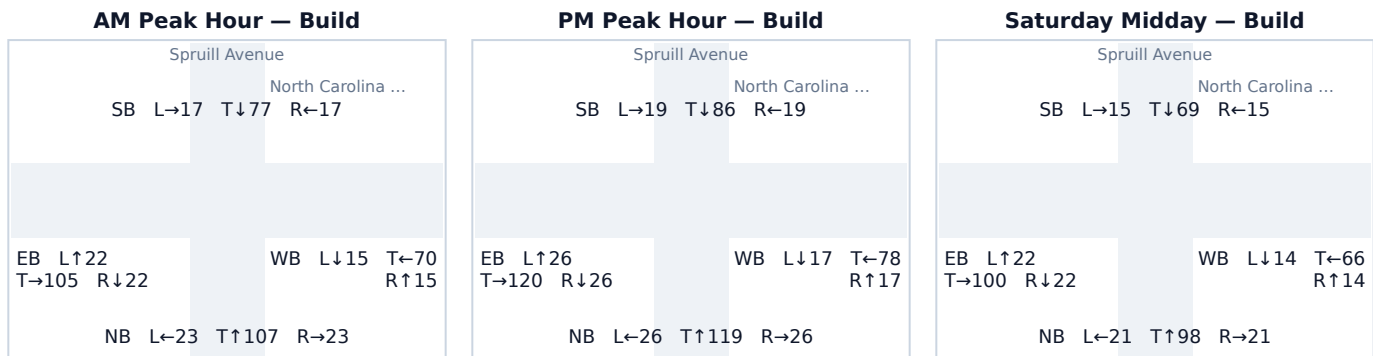
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Spruill Avenue & North Carolina Avenue

Signal ID	charleston-sc-753
Location	Central Charleston (SC) · 0.60 mi from site
Added PM peak trips	21
Existing / No-Build (PM)	LOS B · 16.8 s/veh · v/c 0.31
Build (PM)	LOS B · 17.0 s/veh · v/c 0.32
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	167	4	171	0.21 → 0.21	15.6 → 15.6	B → B	107
SB	123	0	124	0.15 → 0.15	15.0 → 15.0	B → B	75
EB	161	11	172	0.20 → 0.21	15.5 → 15.6	B → B	108
WB	106	6	112	0.13 → 0.14	14.8 → 14.9	B → B	68

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	6	29%
WB Through	6	29%
EB Right	4	19%
NB Left	4	19%
EB Left	1	5%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

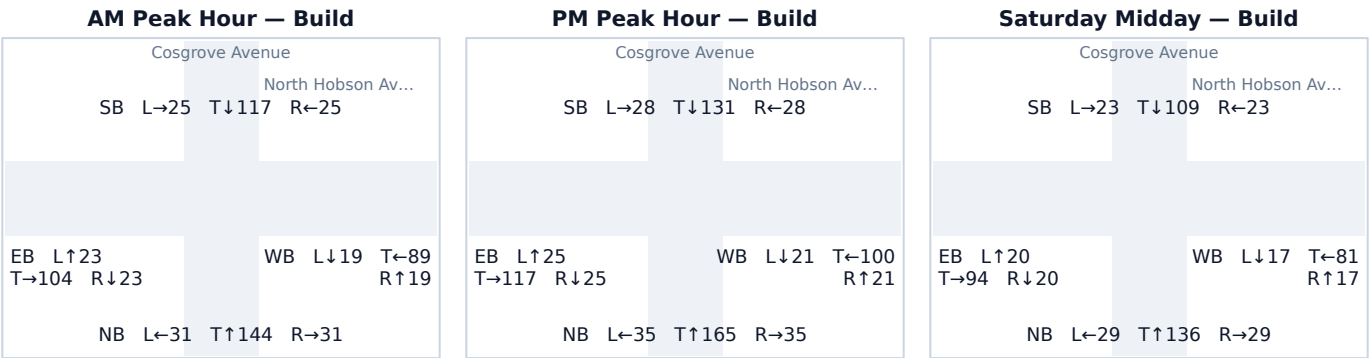
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on

its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Cosgrove Avenue & North Hobson Avenue

Signal ID	charleston-sc-82
Location	NW Charleston (SC) · 0.62 mi from site
Added PM peak trips	20
Existing / No-Build (PM)	LOS B · 18.0 s/veh · v/c 0.39
Build (PM)	LOS B · 18.2 s/veh · v/c 0.41
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	225	11	235	0.28 → 0.29	16.4 → 16.6	B → B	153
SB	180	7	187	0.22 → 0.23	15.8 → 15.9	B → B	118
EB	167	0	167	0.21 → 0.21	15.6 → 15.6	B → B	104
WB	139	2	142	0.17 → 0.17	15.2 → 15.2	B → B	87

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	8	40%
SB Through	7	35%
NB Right	3	15%
WB Left	2	10%

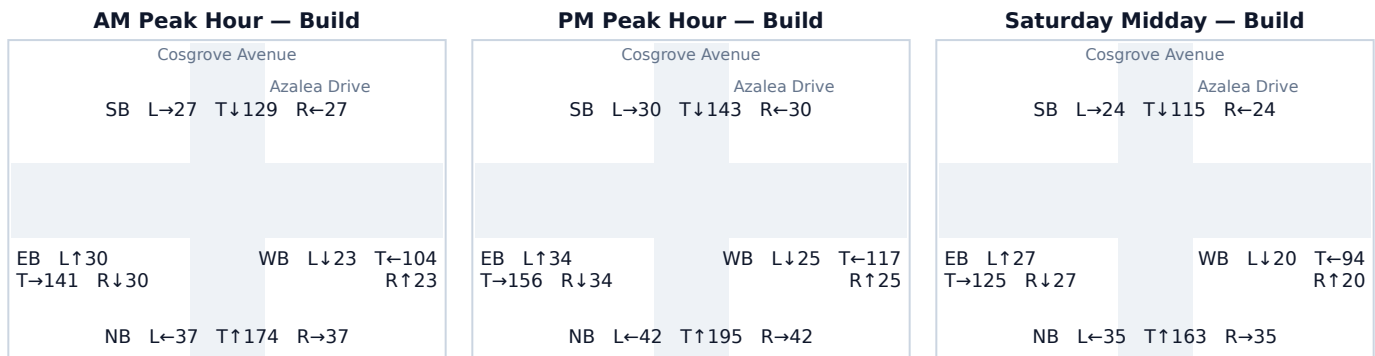
Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Cosgrove Avenue & Azalea Drive

Signal ID	charleston-sc-130
Location	Central Charleston (SC) · 0.67 mi from site
Added PM peak trips	14
Existing / No-Build (PM)	LOS B · 19.3 s/veh · v/c 0.48
Build (PM)	LOS B · 19.5 s/veh · v/c 0.48
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	265	14	279	0.33 → 0.34	17.0 → 17.3	B → B	187
SB	203	0	203	0.25 → 0.25	16.1 → 16.1	B → B	130
EB	224	0	224	0.28 → 0.28	16.4 → 16.4	B → B	145
WB	167	0	167	0.21 → 0.21	15.6 → 15.6	B → B	105

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	14	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.